

Integrating Multiple Models For Effective Video Retrieval and Multi-stage Search

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ABSTRACT

Video is one of the most prevalent forms of data due to the widespread availability of recording devices. This makes video retrieval systems essential since they assist in locating a video segment within a dataset that most closely matches a given query. One of the difficulties of video querying is the processing of multimedia data (images, audio, and text). In addition, it is important to integrate temporal information, as inquiries frequently pertain to the depiction of events occurring within a specific time frame. Thus, this study introduces an innovative system capable of not only integrating various types of models but also effectively managing temporal searches through multi-stage processes. The efficacy of the systems was demonstrated at the AI Challenge 2023, which took place in Ho Chi Minh City, where our team got the best accuracy across all other contestants in the qualifying phase and received the top 1 position among the 60 participating teams.

CCS CONCEPTS

• Information systems → Search interfaces; • Human-centered computing → Interactive systems and tools.

KEYWORDS

Lifelog Event Retrieval, Video Retrieval, Content-Based Image Retrieval

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1 INTRODUCTION

In the digital age, the explosive growth of video data has become a phenomenon of unprecedented proportions, reshaping the way we communicate, entertain, work, and even secure our surroundings. This makes the demands for retrieval videos increasingly dramatic and exerts significant pressure on current systems.

Inspired by major and rich historical competitions such as Lifelog Search Challenge (LSC) and Video Browser Showdown (VBS)[4, 11, 17], the Ho Chi Minh AI Challenge 2023 [3] provides a platform where visionaries, innovators, and data enthusiasts converge to push the boundaries of what would happen in the realm of video retrieval. This competition allows participants all around the world to embark on a journey that will test their skills, creativity, and problem-solving abilities.

The Ho Chi Minh AI Challenge 2023¹ focuses on the "Video Event Retrieval" track, where participants develop innovative methods to retrieve specific events or actions from video data. This challenge aims to improve the search, identification, and extraction of relevant information from videos by focusing on predefined events or criteria. The objective is to locate videos depicting specific events described in predetermined formats or criteria. Participants should work on improving techniques related to video analysis, content understanding, and event recognition, striving for accurate and reliable video event retrieval results. This year's competition was almost like the year before using two types of query of Known-Item Search (KIS):

- **Textual KIS:** Participants will be provided with a description of a specific video segment within the database, and the task is to accurately locate that particular video segment.
- **Visual KIS:** A single video clip (about 20 seconds long) is randomly selected from the dataset and visually presented on-site. Participants need to find one exact frame with the condition that the frame must be in the shown clip.

¹<https://aichallenge.hochiminhcity.gov.vn/>

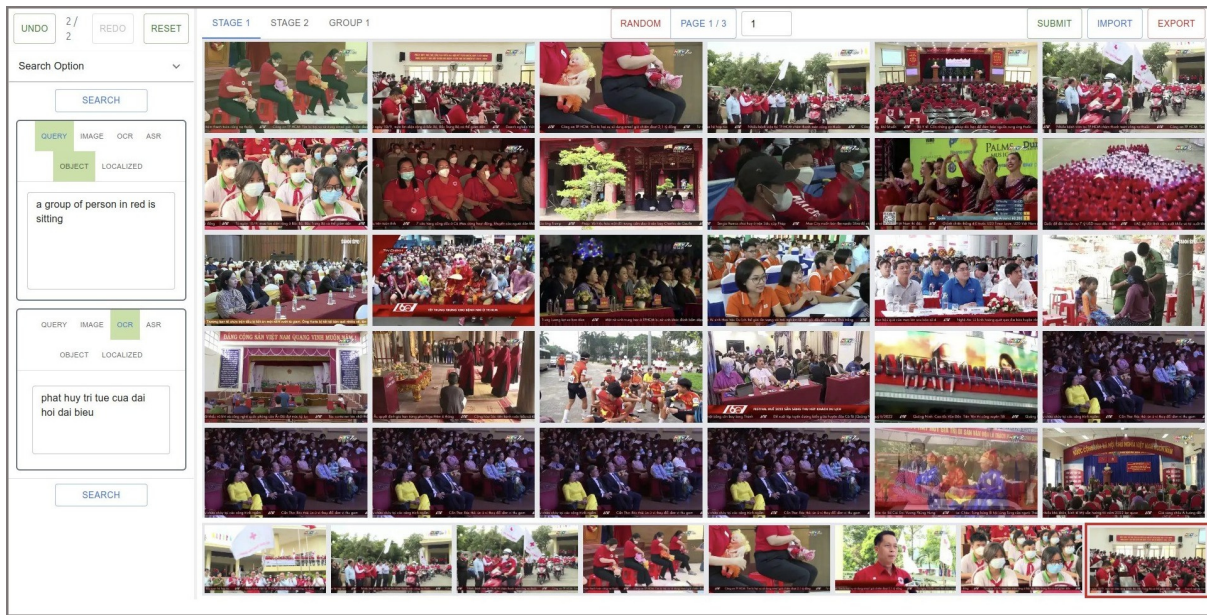


Figure 1: The main page of our system.

This paper introduces an efficient system capable of conducting multi-stage search across multiple models, encompassing text-image search via OCR through PARSeq[1], ASR utilizing Whisper [15], BEiT-3[20] for Free-text, Image or Localized Search, Object Detection using Co-DETR[23]. The multi-stage search involves pinpointing pertinent information within defined time intervals, spanning diverse data formats like text, images, audio, and video. Our work showcases a comprehensive system facilitating both multi-stage and multimodal search, seamlessly integrating state-of-the-art techniques in video retrieval across various stages.

Our system’s effectiveness was demonstrated by the competition’s results, where we proudly secured the 1st rank on the public leaderboard.

2 RELATED WORKS

In recent years, many teams have participated in LSC and VBS contests. Prominent systems among them are [4, 11, 17]:

Vibro[5] used a ViT-L[2] model which is pre-trained using CLIP[14] and fine-tuned with a blend of some public datasets of images like ImageNet21k[16] and Google Landmarks v2[21]. This intricate approach effectively fused text and image data into a shared common space, elevating **Vibro**’s capabilities in retrieving relevant content from images, showcasing the sophistication of advanced techniques in image-related tasks. Additionally, Vibro also incorporates a “2D-sorted map” (a self-organizing-map (SOM)[8]) to display retrieved results.

VideoCLIP[12] used the CLIP[14] model for Rich Text Query, used the query-by-image feature to find all visually similar frames from the selected one, and exploited Google Vision API to extract any text visible in the images. Moreover, VideoCLIP also supports

searching by top dominant colors (12 colors in total in their system) similar to VideoFall[13] with 32 colors, for example: sky was dominantly blue, or forest was dominantly green.

In addition, many teams that participated in these competitions also utilized temporal search, with different ways of combining scores from multiple stages. Some teams have competed with others by building a system that is intuitive, simple, and easy to use but effective such as V-FIRST[19] at VBS[11], and LifeLens[6] at LSC[17].

Having drawn out takeaways from all the ideas of previous teams, we built our system with the ability to take advantage of those methods and integrate them into each stage of the temporal search technique, thereby making the searching process more effective, therefore, uses the strengths of each method and reduces the search space we need to look for. Simultaneously, the system is also developed with user experience as the top priority in mind, thereby making search operations faster and simpler.

3 DATASET

3.1 Introduction

In this year’s AI Challenge held in Ho Chi Minh City, teams are required to find specific video segments from a dataset of news videos encompassing various domains. Since the number of videos is 1290, which totals 360.95 hours (see Tab. 1) and if the minimum framerate were 25, this could result in 32.5 million frames. The exhaustive search would be infeasible. Therefore, we came up with a solution to gather the necessary frames for our system.

Moreover, each video within the dataset is accompanied by annotations containing information about the metadata such as date, channel, and headlines. These annotations provide valuable information that can be used to facilitate targeted video searches.

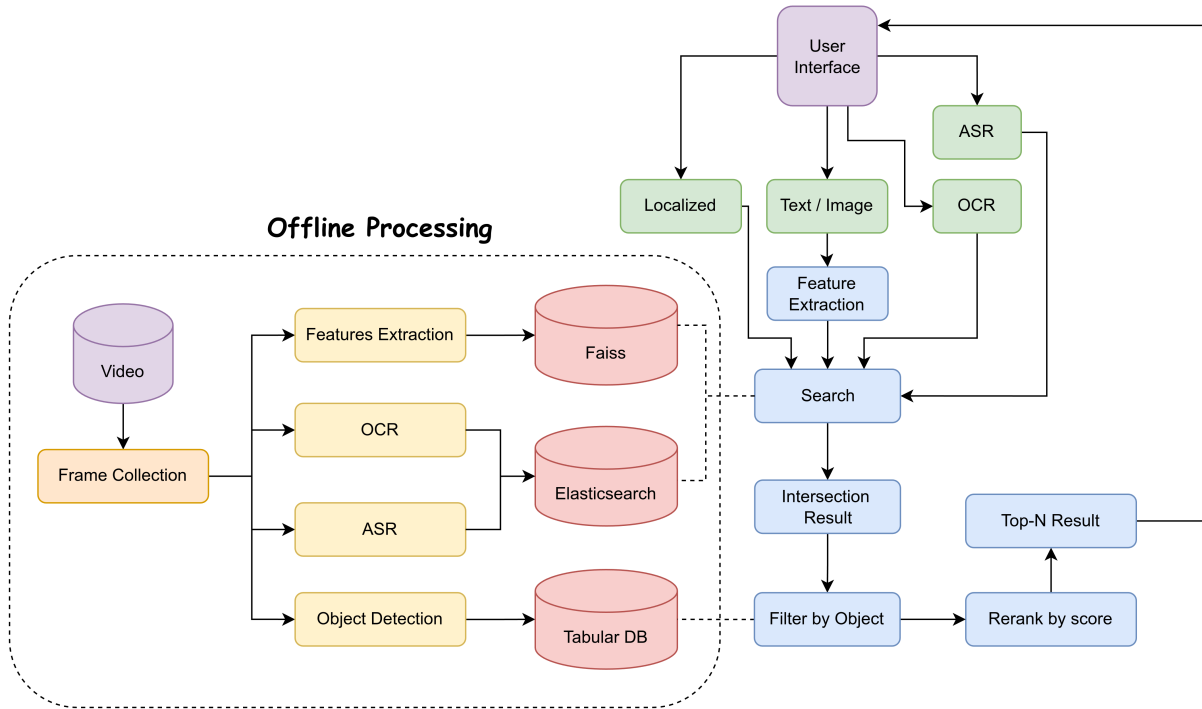


Figure 2: The overview Pipeline of our Multimodal Multi-stage System.

Table 1: Dataset detail

	# Videos	# Hours
Batch 1	299	98.83
Batch 2	438	116.68
Batch 3	473	145.44
Total	1210	360.95

3.2 Frame collection

To gather all essential frames, we employed a keyframe selection process based on the following fundamental concept: if the frames under consideration exhibit similarity with a specific preceding frame or with frames already stored in the database beyond a certain threshold, we added those frames into our database. For each N frame, we used a deep learning model to extract the visual feature vectors, and then we compared the feature vectors with the ones that had been in the database. If the difference is below a carefully chosen threshold, the frames being compared will not be stored in our database. By collecting frames in this way, we saved a lot of storage, while also helping to improve system performance and to search more efficiently.

After this step, we ended up with a total of 639,098 frames (a dramatic decline from the original 32.5 million frames). These frames are extremely useful for text-image retrieval tasks, object detection, OCR, and even thumbnails for displaying with User Interface (UI).

4 SYSTEM OVERVIEW

Our system consists of two fundamental components: the front-end part and the back-end part. We used a state-of-the-art model for zero-shot image retrieval BEiT-3, with the ability to retrieve image match at specific positions; text models for text detection and transcribing detected regions; ASR model's advantage was used for searching for frames of which audio is matched with the query, which filters the results by counting objects and positioning them. With the front end, we split the UI into two parts to achieve an efficient and better searching experience, one for inputting query, and the other for showing the results, with the ability to show near frames and frames from other stages with temporal search. For details, a screenshot of the UI is shown in Fig. 1, while the pipeline of our system is shown in Fig. 2.

4.1 Scene Text Recognition

Due to the nature of this competition's dataset i.e. broadcast news, which often contains a lot of headlines, text overlays, ... These texts can provide useful clues about the content and context of the news videos. Therefore, we chose Scene Text Recognition (STR) to make use of this valuable information.

To achieve the best performance, we separated STR into 2 main tasks, text detection and text recognition. This allowed us to utilize the model in each separate task. For text detection, we used DeepSolo[22], a state-of-the-art scene text detection model, to locate the text regions in each frame. Then, we applied a pre-trained text recognition model called PARSeq[1] which is trained on Vietnamese data, to transcribe the text. Our experiment showed that

PARSeq[1] outperformed other Vietnamese STR models such as VietOCR².

Finally, we used Elasticsearch³ with Fuzzy Search to store the results and reranked the frames based on the similarity between the input query and the recognized text. This way, we can avoid important frames lost due to recognition errors in Vietnamese words. The illustration demonstrates the improved search results by incorporating STR is shown in Fig. 6.

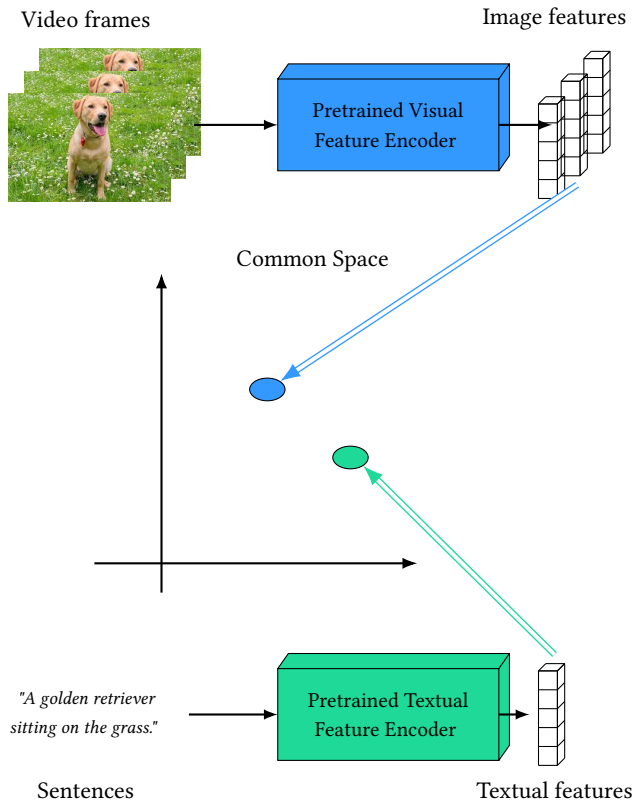


Figure 3: The visual-text embedding used in CLIP[14].

4.2 Automatic Speech Recognition

News videos are a rich data source that can be analyzed in various ways. One of the most important aspects of news videos is the audio content, which contains information such as speech, music, sound effects, and background noise. Analyzing audio content enabled us to develop a unique retrieval method that could match queries with relevant news videos. Even if the queries do not explicitly mention audio-related terms, we can infer the potential audio content from the textual information in the queries, and use that to enhance the retrieval process. This makes the audio information highly valuable for our task. To extract audio information from news videos, we utilized a powerful Automatic Speech Recognition (ASR) model, Whisper[15], a model that can handle noisy and diverse audio inputs, and generate high-quality transcriptions.

²<https://github.com/pbcquoc/vietocr>

³<https://www.elastic.co/>

By using Whisper[15], we could obtain rich and accurate textual representations of the audio content in news videos.

For each frame, we took a segment of audio that is N seconds long, centered around the frame's time. Then we use the Whisper[15] to figure out the sound in that segment. After that, we used that audio content to represent the frame, and then we could search for that frame later, just like what we did with OCR. We also used Elasticsearch's Fuzzy Search to make our search more flexible and accurate.

4.3 Free-Text and Image search

Finding images that match a given text is a difficult problem that requires both natural language understanding and computer vision. Many current retrieval systems have tackled this problem by using models with training methods that can learn how to associate image-text pairs. In recent years, CLIP[14] has become a trend in image retrieval-related tasks and is applied to most systems. However, in the system we used for participating, we used BEiT-3[20] which is a vision transformer that leverages Multiway Transformers, masked "language" modeling across modalities, and an effective scaling strategy. This model has shown superior performance on image retrieval tasks compared to other models. We also used BEiT-3[20] to encode the visual features of each video frame that we collected and then applied Faiss[7], which is a fast and efficient tool for indexing and searching, to index them for retrieval. From a text or an image query from a user, we also encoded it using BEiT-3[20] and then searched for the most similar video frames in the indexed database using Euclidean Distance.

4.4 Localized Search

One of the challenges we faced in improving our system for the Visual KIS task was how to enable users to search for specific positions in the video content that are relevant to their text queries.

To address this issue, we implemented a feature that ensures the content similarity between the text query and the video frames is measured at the level of particular positions within each frame. Our approach was to divide every frame into a grid of $M \times N$ cells and treat each cell as a separate frame. Then, we used a visual feature extraction method to obtain the features of each cell and stored them in a database. This way, we could perform a text search on the cell-level features and retrieve the most similar positions in the video frames.

With Localized search, we found that there can be multiple results that originate from a single frame potentially leading to duplicate results. However, we only need to keep the original frame to display on the system. To achieve this, we calculate the mean score for the original frame that we divided into smaller cells in the previous step. This means scores have the ability to cooperate and interact with other modules of our systems. With the goal of creating a multimodal system, our approach enables us to retrieve the frame with specific locations that best match our query.



Figure 4: Filter object by position helps us search more efficiently.

4.5 Object Detection

With the rise of vision-language models, such as those proposed by [20], have demonstrated the great potential for image-text retrieval systems [5]. However, these models may encounter challenges when locating small objects or when the presence of multiple objects in a frame leads to confusing retrieval results. To address this issue, incorporating object information, such as object class and location (see Fig. 4), can significantly enhance precision. This technique is commonly known as "object filtering." By employing object filtering, we can selectively choose results that specifically contain the object of interest, thereby improving the accuracy and relevance of the retrieval process.

In the current year's competition, we leverage Co-DETR [23] with the Swin-L [10] backbone. This model is first pre-trained on Objects365 [18] and then fine-tuned on COCO [9], which offers a diverse range of categories and encompasses common objects found in everyday life. Consequently, this backbone is ideally suited for our system's needs.

For each frame, we will store the count of objects per class and the corresponding bounding box coordinates. In this way, the system can efficiently filter out irrelevant results and focus on the ones that match the user's query.

4.6 Temporal Search

As a key feature of our system, we introduce a multi-modal multi-stage search method, distinguishing it from traditional systems that typically employ a two-stage approach with limited feature combinations, often restricted to text query. In our approach, each stage integrates diverse features, including Free-text search, Image search, Localized search, OCR, ASR, and filtering by Object. This integration substantially improves the efficiency of the search process, especially for Visual KIS query, allowing us to comprehensively analyze every stage of the video.

Certainly, our algorithm to calculate scores for results from temporal search is based on Vibro [5]. In basic concept, for each frame at the current stage result, we take all frames at the next stage result in the same video, satisfying the condition about time, similarity to subtraction of two frames. Subsequently, we combine the frame current result with each of the found frames and select the pair with the highest sum score. With that pair, we take the last frame to continue the same process. As a result, we obtain a set of sequences of frames re-ranked from top to bottom based on the score. The UI of the system where users input the query for temporal search, is shown in Fig. 5.

4.7 User Interface

We also designed a system that is easy and flexible for users to use, in order to enhance their search experience. The user interface is

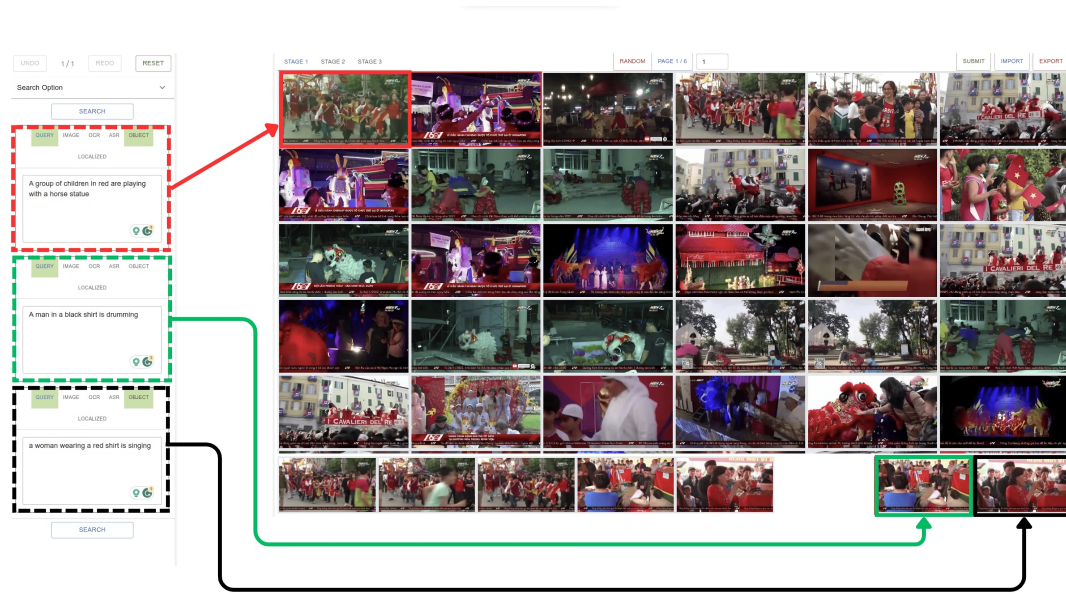


Figure 5: The UI of our system when using the Temporal search.

designed with only two main parts, query, and results, as illustrated in Fig. 1.

- **Query:** In this part, users can input any query to retrieve the frame they are interested in, including Free-text search, Image search, Localized search, OCR, ASR, and filter by Object. Each of them is in a Tab, so the users can easily change between them.
- **Results:** This part shows users the results. Users can see the top 3000 frames in either a map that is sorted in 2D, based on how relevant they are to the current query. Besides that, when the user clicks on any frame, our system also shows nearly frames with the clicked frame, when using temporal search, the system also has the ability to display frames that satisfy each stage, which helps users have an overview of the selected frame, quickly check them out and see how they match the query and decide what to do, thereby helping to search more effectively.

5 RESULTS

Through meticulous preparation and collaborative teamwork, our team utilized all the features that we presented to achieve a very good result in the preliminary round of the contest despite the fact that the public leaderboard is only based on 50% of the ground truth (the other 50% is hidden from contestants), but this can prove that our efficient system. From the ranking (see Fig. 7), our system has achieved top 1 in the competition with nearly 60 participating teams, the results show that our system has better performance than other teams.

6 CONCLUSION

In this paper, we employed a system with leverages many models such as free-text search, scene text recognition, audio recognition,

filter by object, and search with position attention. Besides, with complex queries, our system also allows combining models and expanding time queries easily. Our friendly and easy-to-use system has shown a significant result in the qualification round.

Through the strategic application of these advanced methodologies, we have not only adeptly addressed a diverse spectrum of query demands but have also substantially enhanced overall user experience. Our relentless dedication has culminated in the development of an astute and versatile query system that boasts formidable prowess in its functionalities, all while maintaining an inherent user-friendliness and an effortlessly navigable interface. This synergy guarantees a flawlessly harmonious and gratifying interaction for all categories of users.

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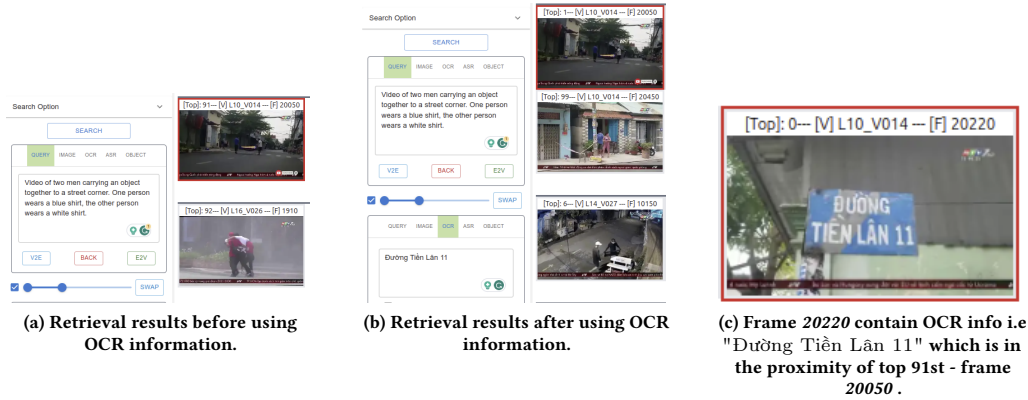


Figure 6: The illustration demonstrates the improved search results by incorporating information about the text present in the image. In the left image - a), the search result was ranked at the 91st position. In the middle image - b), the search result was re-ranked with the additional OCR information, and as seen, the query result was significantly enhanced, reaching the top 1 position.

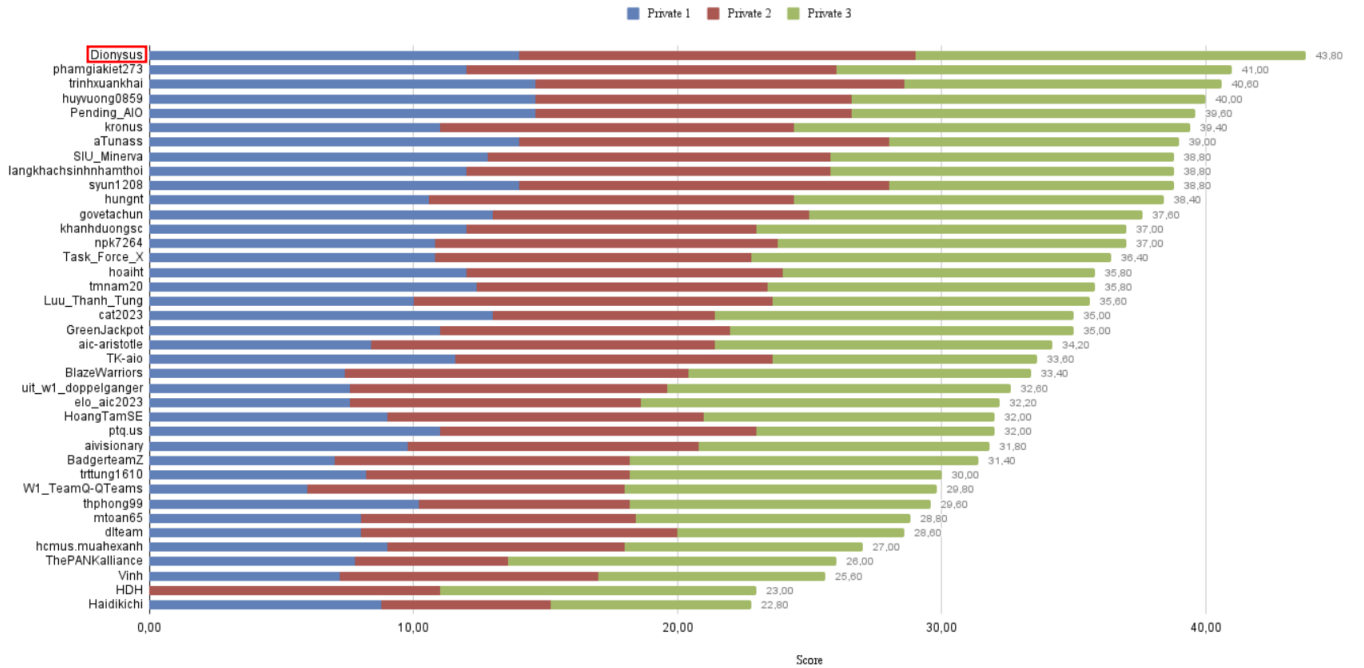


Figure 7: Our performance on public tests, about 50% of scores in the elimination round. Our system achieved 43.8 out of 45 and took first place.

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